Fieldnote

Source-linked operational answers, with shared facts and private conversational memory.

Data and retrieval

The released snapshot contains 600 wells, 1,25,198 inspections and 17,415 maintenance records. Node.js 24 and SQLite/FTS5 retain original JSON/JSONL records, linked free-text reports, manuals, SOPs and new letters/MOC/recall documents. Newer releases supersede changed record IDs while retaining earlier versions. Type and team history, late-entered dates, cost revisions and structured corrections are applied.

Agentic layer and deterministic tools

Gemma 4 proposes a schema-constrained query plan. Literal scope, dates and operations are checked; numerical results come from linked data tools. Tools handle inspection status, gas-response deadlines, bypass timelines and memo extensions, laboratory evidence, current pump-lot inventory, cold-weather checks, reported delays and temperature comparisons. Unknown reservoir pressure, oil output and incident-specific H2S readings are not invented. Risk ranking is explicitly a heuristic judgement grounded in operator quotes and warning signs.

Memory, feedback and provenance

Each user has persistent private conversations, topic/entity sets and display preferences. “First question” returns to that session's first result set even after unrelated topics. Shared replacement reports retain the speaker, effective date, affected installation and verification state. Direct responsible-user reports are distinguished from hearsay; missing work orders remain visible. Preferences such as sorting and currency units apply only to their author. Corrections preserve originals and recompute subsequent answers.

Interface and evidence

Questions and answers are shown in a compact chat interface with cards, tables, charts and maps. Sources, complete paginated data and technical traces open on demand. Every final turn was typed through the actual browser UI under the named user/session; full response JSON and screenshots were saved. Unsupported or malformed model output is rejected, with a labeled local fallback where supported.

Models, costs and boundaries

Model: gemma-4-26b-a4b-it via Google's Gemini API. This final browser replay had 18 model-backed read turns, 3 local memory/feedback writes, and 0 provider fallbacks. Provider-reported usage: 16,392 input and 311 output tokens; failed requests may lack usage. The supplied guide describes Gemma as free-tier; invoice cost was not measured. The result is a hybrid application result, not unguarded model accuracy.

Next: broader paraphrase evaluation, stronger semantic checks for novel SQL, authenticated deployment and richer event-time memory. AI-assisted development using OpenCode; participant review and an own-voice demo are required. Application: MIT. Dataset and fictional documents: ©2026 FlytBase, MIT.

Measured final results

All 21 turns of the 11 final questions completed in the browser, in order and with the specified users/sessions. Source/rule checks passed. No official answer key was supplied for the judgement questions; those conclusions remain explicitly qualified.

Q1	Not derivable: no reservoir-pressure measurements; ratings and flowline tests are not reservoir pressure.
Q2	Not derivable: no metered oil-production volumes; temperature and spills cannot rank production.
Q3	26 gas reports; 17 met and 9 missed. January uses the 12-hour rule; earlier reports use 24 hours.
Q4	3 current bypasses; 6 current/historical exception rows. MOC-031 extension applied through MOC-044.
Q5	6 spills covered by the letter; 3 lack the required lab-sample sentence.
Q6	15 status disagreements; latest eligible note/inspection and structured corrections resolve current state; later restarts remain unconfirmed.
Q7	10 evidence-backed candidates; first three: WELL-0219, WELL-0332, WELL-0336. Ranking is a transparent triage judgement, not a calibrated 90-day failure probability.
Q8	Nathan's own-well replacement report is shared and flagged pending paperwork; Olivia's hearsay does not remove another pump. Private sorting/currency preferences remain Nathan-only. 8 documented recalls shown to Robert; 3 in Nathan's team.
Q9	First recall set survives topic changes. Nathan Pierce leads 150 Northwest wells. H2S at WELL-0332 is not recorded. 2 recalled wells are not producing; first priority: WELL-0332.
Q10	474 jobs have an explicit report statement quantifying extra shut-in waiting. Shares are within each area's documented delayed jobs; typical means median extra days, with means also shown. 709 other corrective jobs lack this explicit waiting statement and are not treated as zero delay. Southeast Block 31 has the highest area median (10 extra days). This is a descriptive comparison, not proof of an area's operational quality; equipment mix and reporting coverage can differ.
Q11	Assuming “hotter” means operational wellhead temperatures, I compared 2026-01-01 to 2026-12-31, using producing readings, each well equally weighted within type/month. Raw wellhead means are highest in Southeast; raw heat mixes weather, lift type and operation. After subtracting ambient and equally weighting the 36 type/month strata present in every region, Southwest has the highest mean delta, but is hottest in 17/36 comparisons. There is no region that is universally hottest across all matched type/month comparisons. This does not establish reservoir heat or oil output.

What worked and what did not

Temporal joins, changed deadlines, shared claims versus private preferences, and first-topic recall were verified. Missing measurements were reported as unavailable. The dataset does not identify every installed pump lot, so recall counts describe documented affected inventory, not proof that all other pumps are unaffected. Risk scores are not calibrated probabilities; temperature conclusions depend on stated normalization. Google returned intermittent HTTP 500s during development; this replay records the actual fallback count above.

Full answers, every result row and visuals: **results/final-transcript.html**. Raw browser outputs: **results/final-browser-evaluation.json**. Screenshots: **results/final-browser/**. Participant must add the public repository URL and submit the own-voice video through OrganizerHQ.